Enterprise-Style Endpoint Monitoring & Detection Lab

Using Sysmon + Splunk SIEM

1. Executive Summary

This project simulates a real-world Security Operations Center (SOC) environment by implementing endpoint log monitoring, detection engineering, alert configuration, and dashboard visualization using Sysmon and Splunk SIEM.

The objective of this lab was to detect and monitor simulated attack scenarios from a blue-team perspective. Two attack simulations were performed:

1. Brute Force Login Attempts (Windows Event ID 4625)
2. PowerShell Execution Monitoring (Sysmon Event ID 1)

Detection rules were engineered using Splunk Search Processing Language (SPL), alerts were configured for real-time monitoring, and a SOC dashboard was built to visualize security telemetry.

This project demonstrates foundational SOC analyst skills including log analysis, detection logic creation, alerting mechanisms, and threat monitoring aligned with MITRE ATT&CK.

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2. Project Objectives

- Ingest Windows endpoint logs into Splunk SIEM
- Install and configure Sysmon for enhanced telemetry
- Simulate security attack scenarios
- Create detection queries using SPL
- Configure scheduled alerts
- Build monitoring dashboards
- Map detections to MITRE ATT&CK framework
- Document SOC workflow and findings

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3. Lab Environment Architecture

3.1 Environment Components

- Windows 10 Virtual Machine (Endpoint)
- Sysmon installed for enhanced logging
- Splunk Enterprise (SIEM Platform)
- Splunk Forwarder for log forwarding

3.2 Log Flow Architecture

Windows Endpoint

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Sysmon + Windows Logs

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Splunk Forwarder

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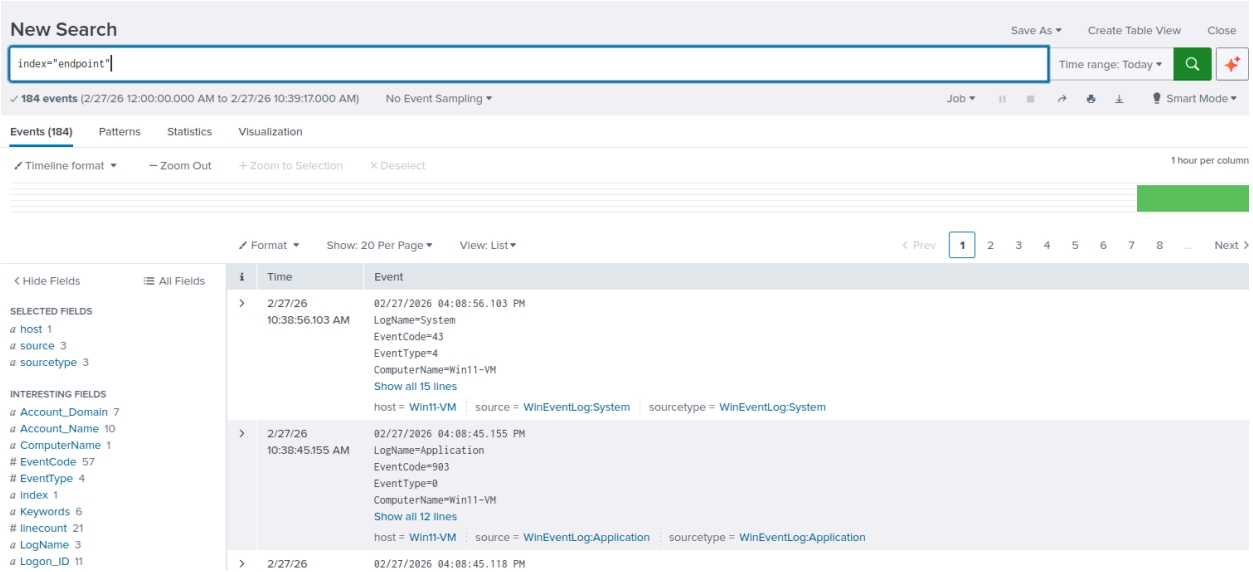
Splunk SIEM (index=endpoint)



4. Log Sources and Telemetry

4.1 Sysmon Event ID 1 – Process Creation

Generated whenever a new process is created on the endpoint.

Important Fields:

- Image (Process Name)
- ParentImage
- CommandLine
- User
- Host

- Process ID

Security Relevance:

Used to detect suspicious execution such as:

- PowerShell abuse
- Malware execution
- Living-off-the-Land techniques

MITRE ATT&CK Mapping:
T1059.001 — PowerShell

4.2 Windows Security Event ID 4625 – Failed Login

Generated when authentication fails.

Security Relevance:

- Brute force attacks
- Password spraying
- Account enumeration
- Unauthorized login attempts

MITRE ATT&CK Mapping:
T1110 — Brute Force

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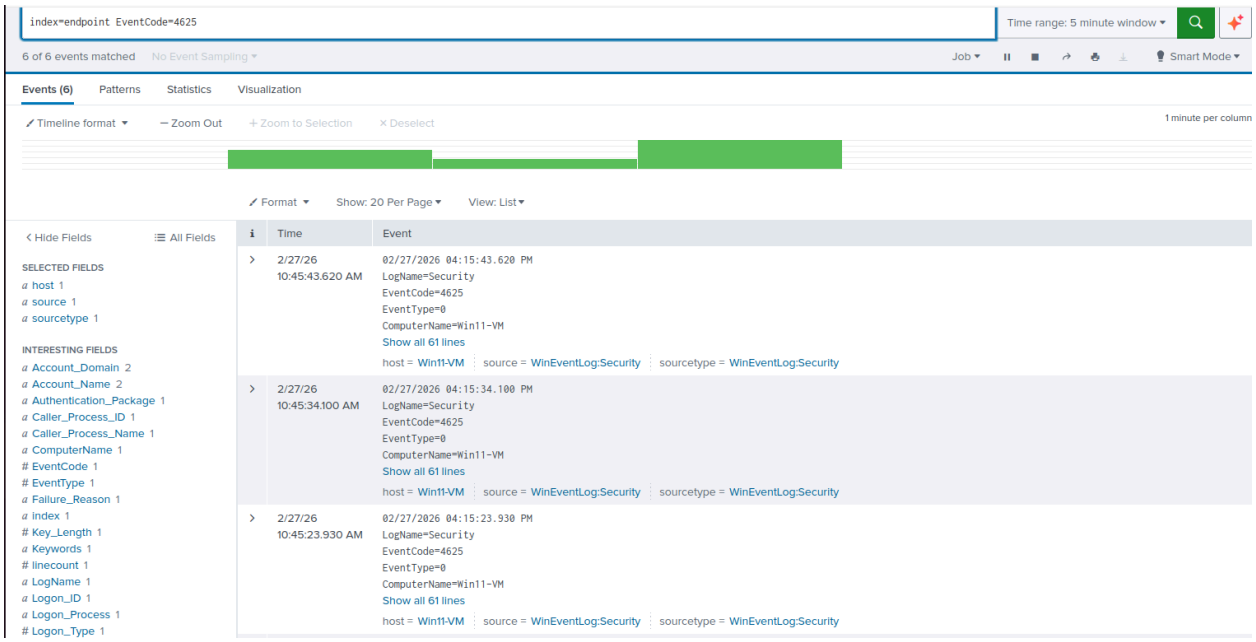
5. Attack Simulations Performed

5.1 Brute Force Attack Simulation

Multiple failed login attempts were generated to simulate authentication abuse.

5.1.1 Log Validation

index=endpoint EventCode=4625

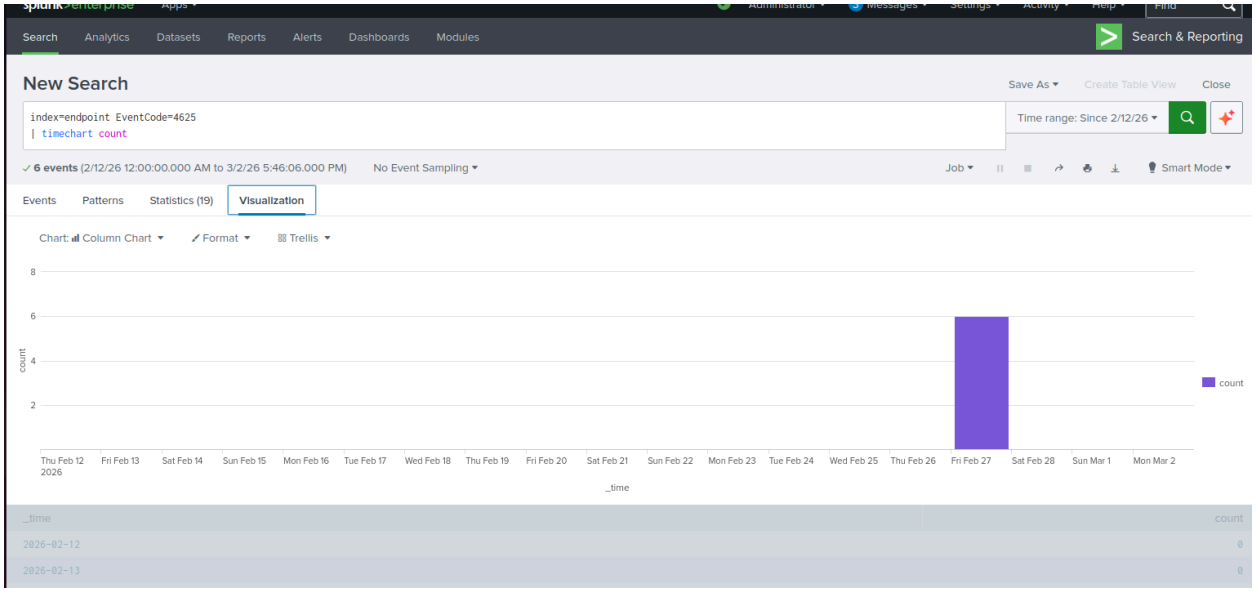


Confirmed that failed login events were ingested successfully.

5.1.2 Detection Engineering

A. Failed Login Trend Monitoring

index=endpoint EventCode=4625| timechart count



Purpose:
Identify spikes in failed login attempts over time.

B. Targeted User Analysis

index=endpoint EventCode=4625

| stats count by TargetUserName

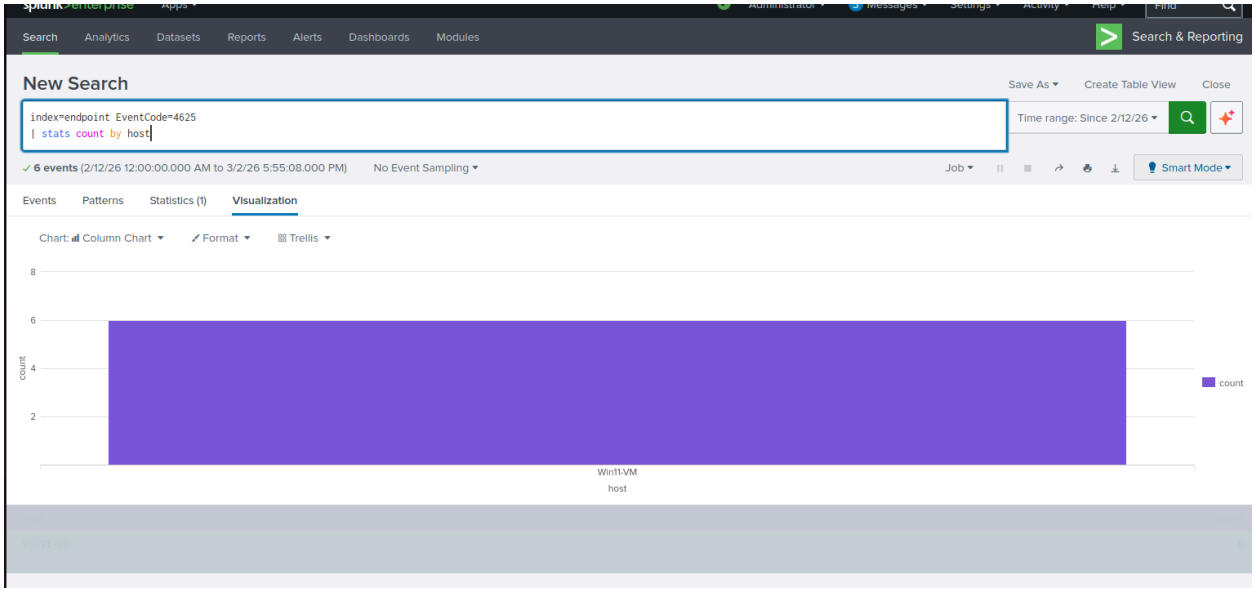
| sort -count

Purpose:
Identify which user accounts are being targeted.

C. Host-Based Analysis

index=endpoint EventCode=4625

| stats count by host



Purpose:
Determine which endpoint is receiving the most failed login attempts.

5.2 PowerShell Execution Monitoring

PowerShell was executed on the endpoint to simulate command execution activity.

5.2.1 Log Validation

index=endpoint EventCode=1

Confirmed Sysmon process creation logs were successfully ingested.

5.2.2 Detection Engineering

A. PowerShell Execution Detection

index=endpoint EventCode=1 Image="*powershell.exe*"

Purpose:
Detect PowerShell execution activity.

B. Process Activity Trend

index=endpoint EventCode=1

| timechart count

Purpose:
Monitor overall process creation trends.

C. Top Executed Processes

index=endpoint EventCode=1

| stats count by Image

| sort -count

| head 10

Purpose:
Identify most frequently executed processes and detect anomalies.

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6. Alert Configuration

Two scheduled alerts were created in Splunk.

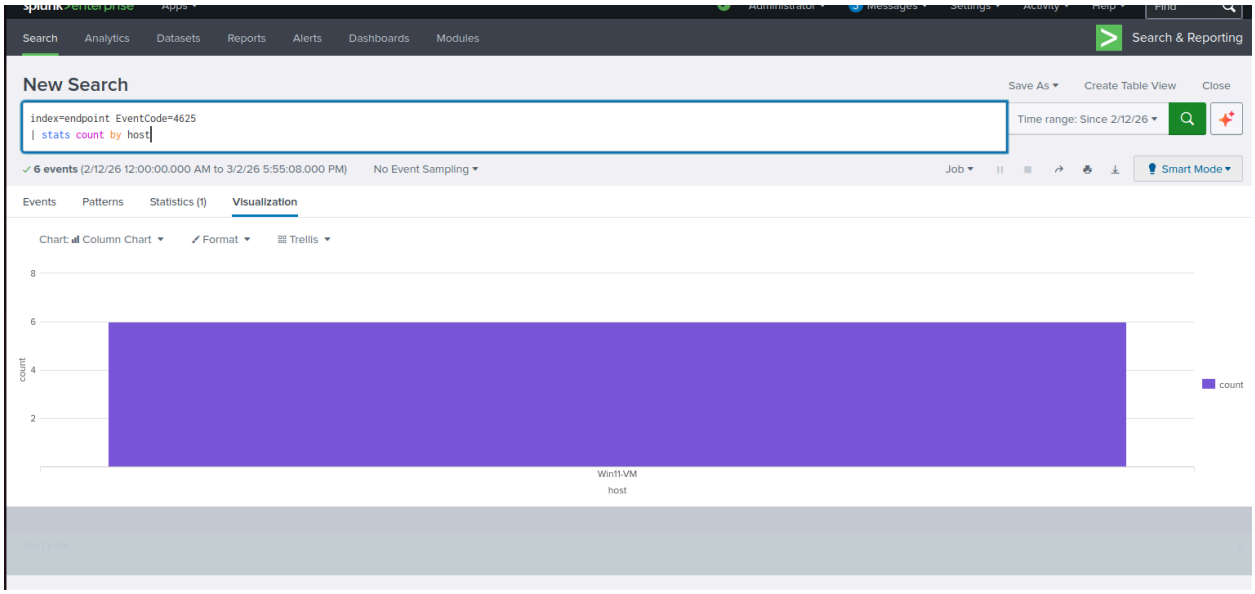
6.1 Brute Force Detection Alert

Title:
SOC-LAB | Brute Force Detection | EventID 4625

Configuration:

- Alert Type: Scheduled

- Run Every: 5 minutes
- Time Range: Last 5 minutes
- Trigger Condition: Number of results > 0
- Action: Add to Triggered Alerts



Purpose:
Detect authentication abuse in near real-time.

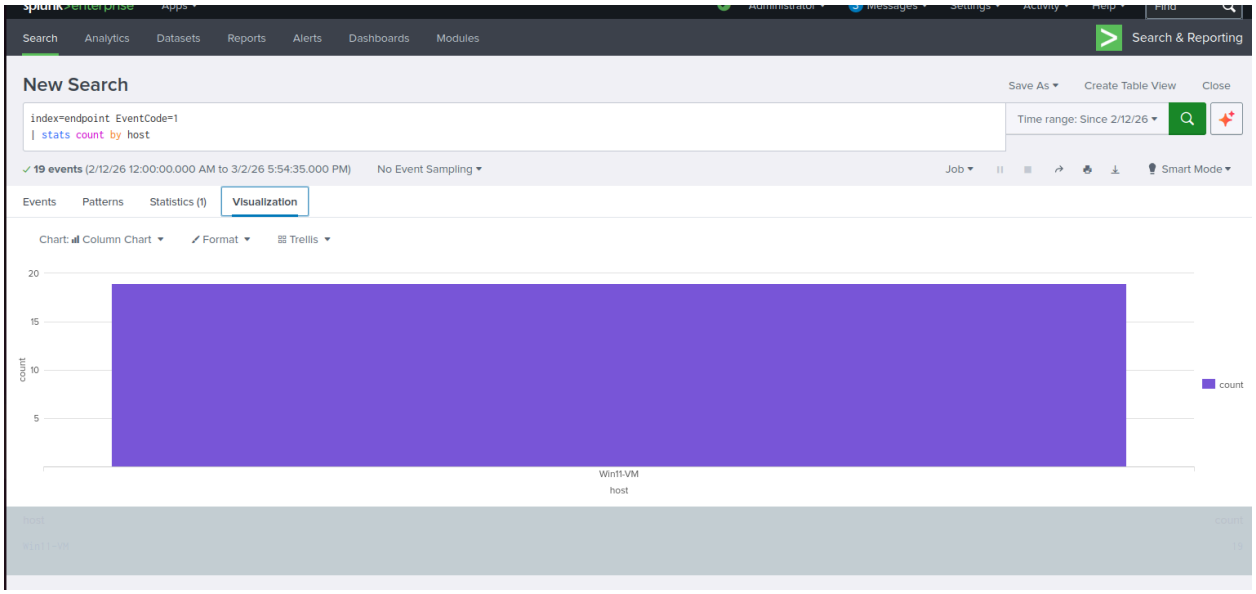
6.2 PowerShell Execution Detection Alert

Title:
SOC-LAB | PowerShell Execution | Sysmon EventID 1

Configuration:

- Alert Type: Scheduled
- Run Every: 5 minutes

- Time Range: Last 5 minutes
- Trigger Condition: Number of results > 0
- Action: Add to Triggered Alerts



Purpose:
Monitor suspicious command execution.

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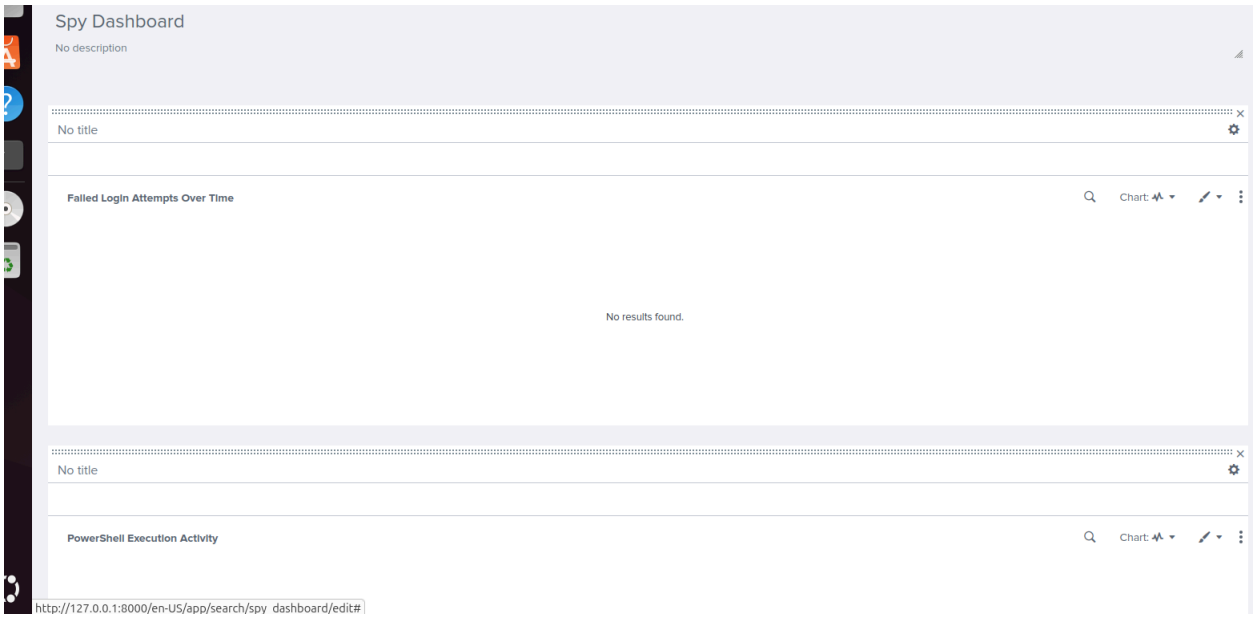
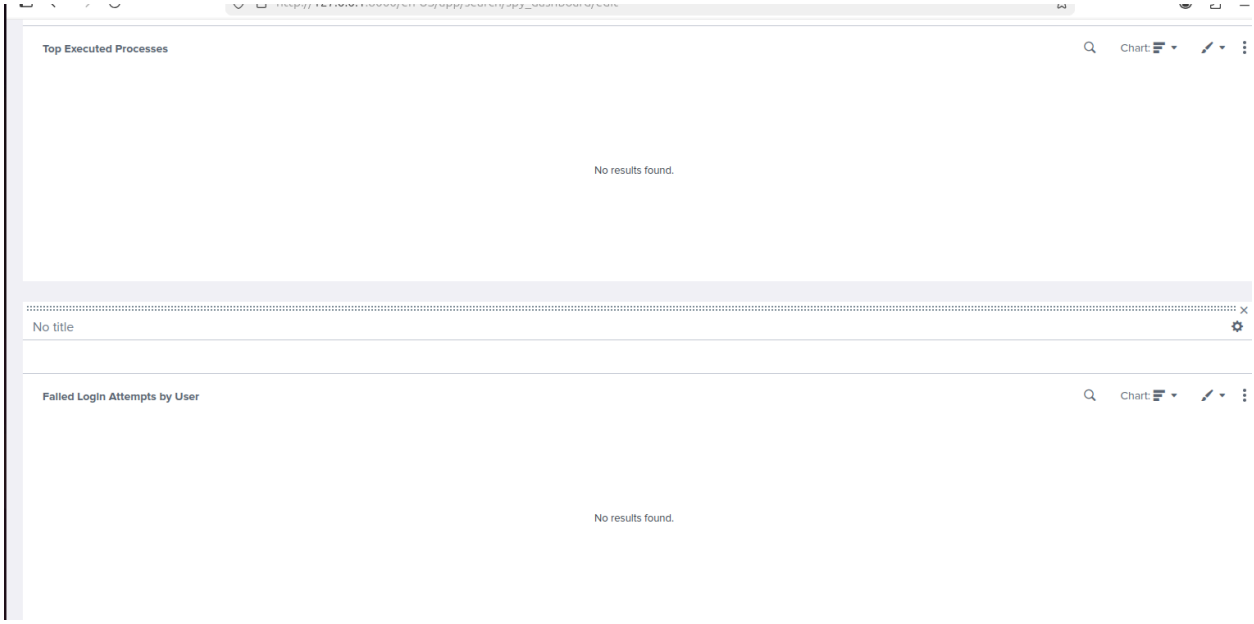
7. SOC Monitoring Dashboard

Dashboard Name:
Endpoint Detection & Monitoring Dashboard

Panels Included:

1. Failed Login Attempts Over Time
2. PowerShell Execution Activity
3. Top Executed Processes
4. Failed Login Attempts by Host





The dashboard provides real-time monitoring capability similar to enterprise SOC environments.

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8. SOC Workflow Demonstrated

This project demonstrates the following SOC lifecycle:

1. Log Collection
2. Log Normalization
3. Detection Engineering
4. Alert Configuration
5. Monitoring & Visualization
6. Investigation Capability

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9. Challenges Faced

- Differentiating between EventCode and EventType
- Field extraction inconsistencies
- Dashboard time range mismatch causing “No Results Found”
- Understanding statistical commands such as `stats count by host`

These challenges improved understanding of Splunk troubleshooting and log analysis.

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10. Learning Outcomes

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- Detection engineering fundamentals
- Dashboard visualization
- MITRE ATT&CK mapping
- Blue-team analytical thinkingThrough this project, I gained hands-on experience in:
- Windows Event Log analysis
- Sysmon telemetry monitoring
- SPL query writing
- SOC alert configuration

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11. Future Improvements

- Implement Sysmon Event ID 3 (Network Connections)
- Add encoded PowerShell detection
- Add threshold-based brute force detection
- Implement correlation searches
- Create incident response playbook

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12. Conclusion

This project successfully simulates a foundational SOC monitoring environment using Sysmon and Splunk SIEM. It demonstrates the ability to ingest endpoint telemetry, engineer detections, configure alerts, and build monitoring dashboards aligned with real-world SOC operations.

The lab establishes a strong foundation for entry-level SOC analyst and blue-team roles by showcasing practical, hands-on detection engineering and log analysis skills.